

ECL101

Basic Electronics Engineering

Winter 2025

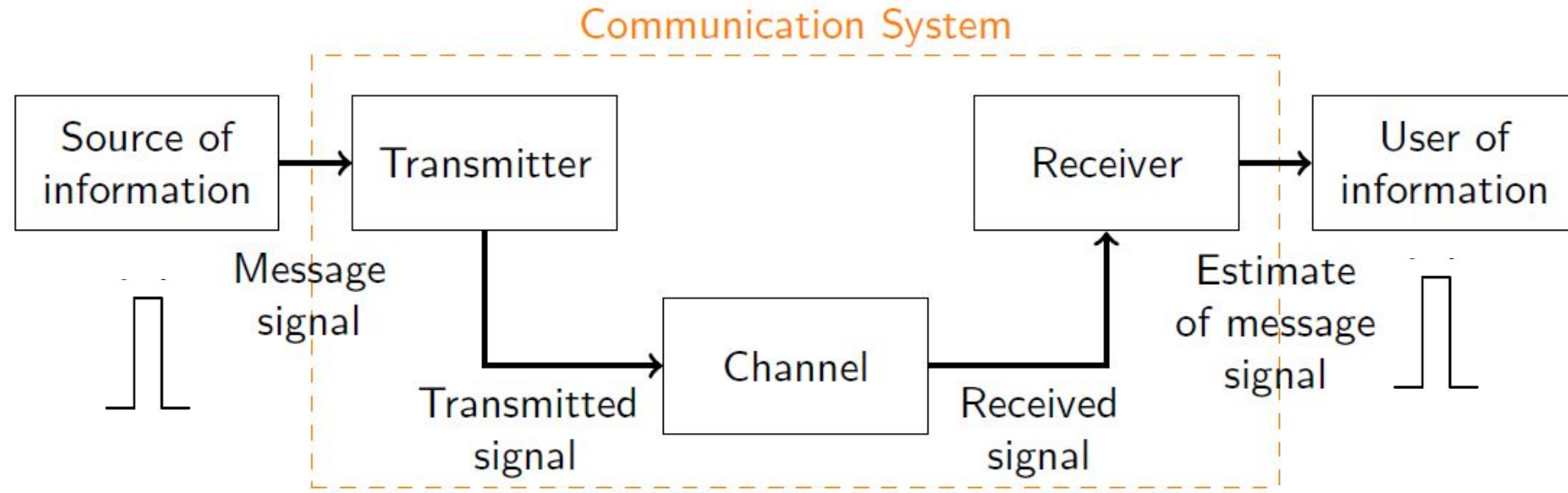


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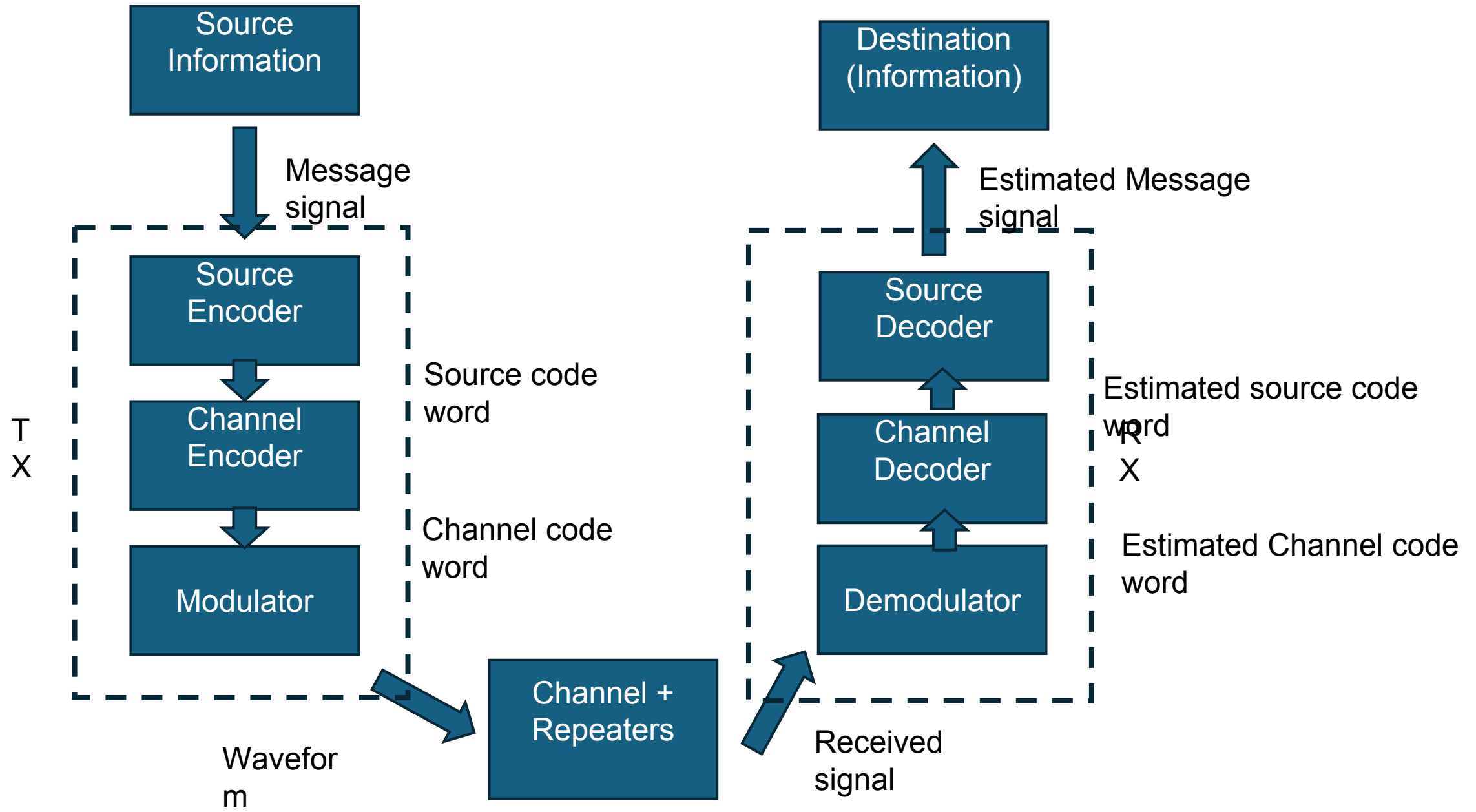
Introduction to Digital communication

Communication system



- ❑ Digital comm. Transmitted signals are a finite set of codewords (bits)
- ❑ Source: Can be Analog signal, like speech (e.g., Voice-over IP)
- ❑ Analog signals are first converted from Analog-to-digital
- ❑ Transmitter encodes to obtain binary bits.
- ❑ Performance is usually expressed as bit-error-rate (BER)

Digital Communication system

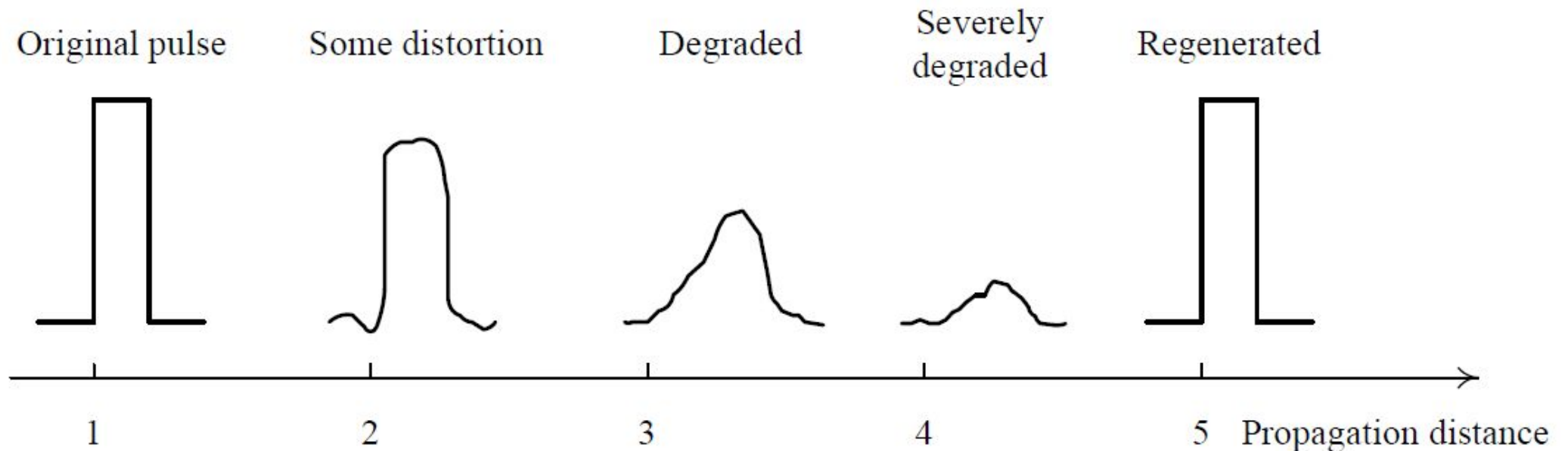


Functions of each block

- ❑ **Source** – The origin of the message (e.g., a computer (digital), microphone (mostly analog)).
- ❑ **Source Encoder** – Converts information into a digital format. (If Analog ->Digitized data!)
- ❑ **Channel Encoder** – Adds redundancy for error detection/correction.
- ❑ **Modulator** – Converts digital data into a suitable waveform for transmission.
- ❑ **Channel** – Medium through which the signal travels (e.g., fiber optics, wireless, telephone lines).
- ❑ **Demodulator** – Recovers the digital signal from the received waveform.
- ❑ **Channel Decoder** – Removes redundant bits and corrects errors.
- ❑ **Source Decoder** – Converts the digital data back into its original form.
- ❑ **Destination** – The final recipient of the transmitted message.

Advantages of Digital communication

- ❑ More Rugged and can withstand more noise and distortion compared to analog communications
- ❑ Viability of regenerative repeaters: The digital signal can be regenerated repeatedly to obtain clean pulses to remove accumulated noise and distortion when traversing through the channel (see Fig. below)



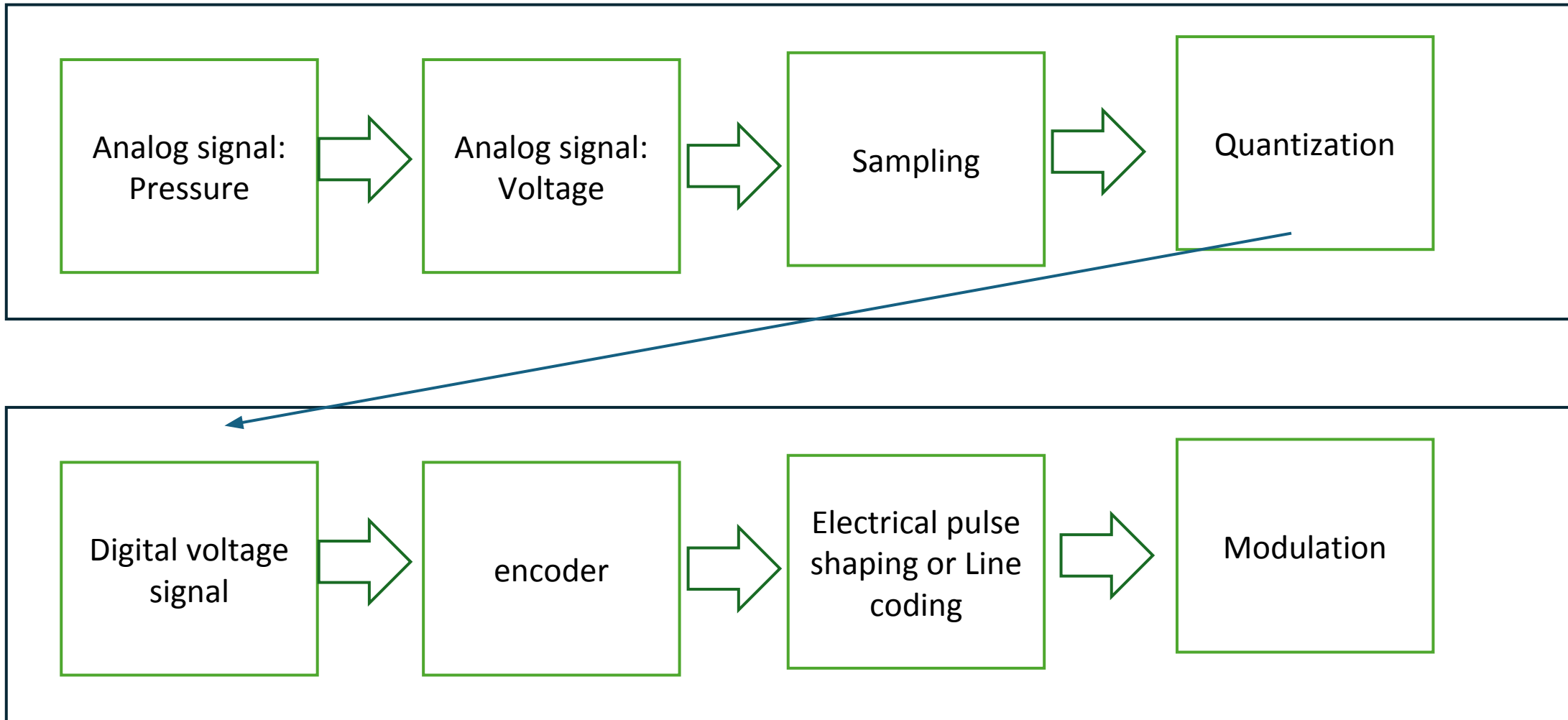
Advantages of Digital communication

- ❑ Digital hardware implementation is flexible and permits use of modern techniques like microprocessors, digital switching, and VLSI
- ❑ Digital messages can be coded to obtain extremely low error rates, high fidelity and allow encryption for privacy.
- ❑ Cost of digital hardware reduces to half the cost every few years while performance improves over time!
- ❑ More efficient in providing exchange of SNR and Bandwidth (see previous slides)

Disadvantages

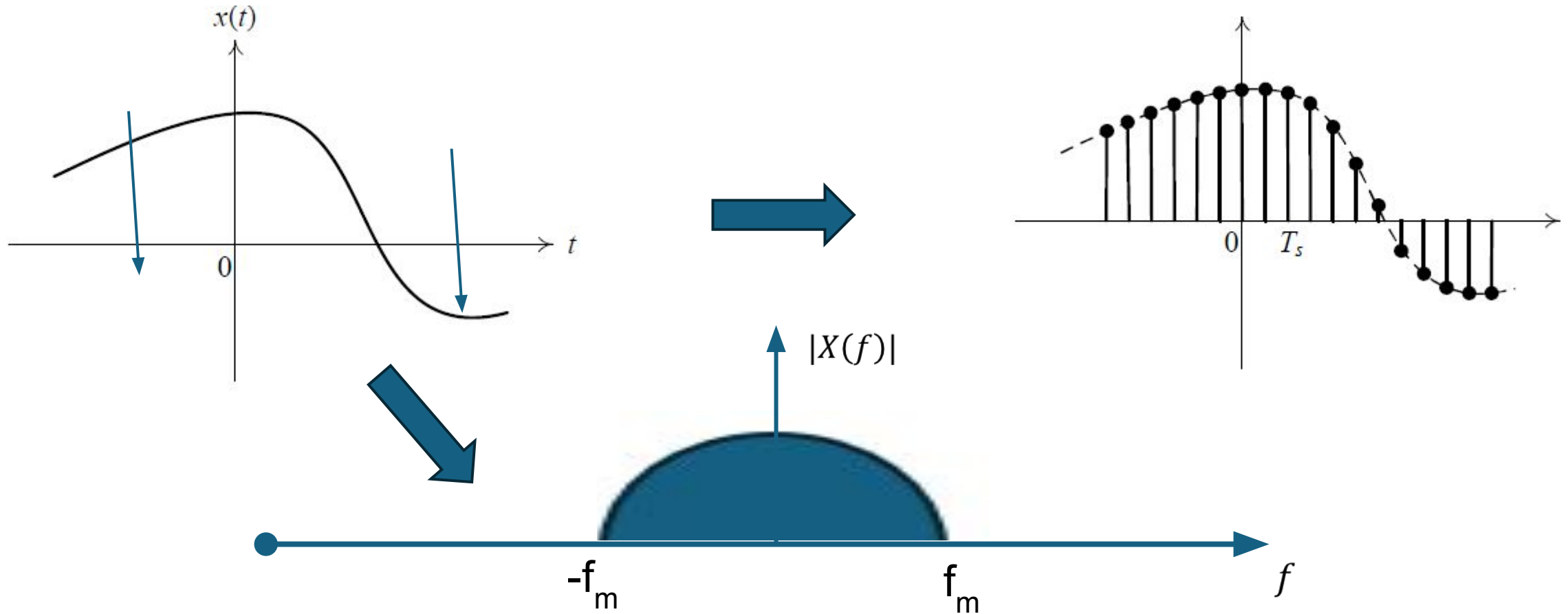
- ❑ Heavy Signal processing is required
- ❑ Synchronization is crucial
- ❑ Larger transmission bandwidth is required

Digital Passband Modulation Process



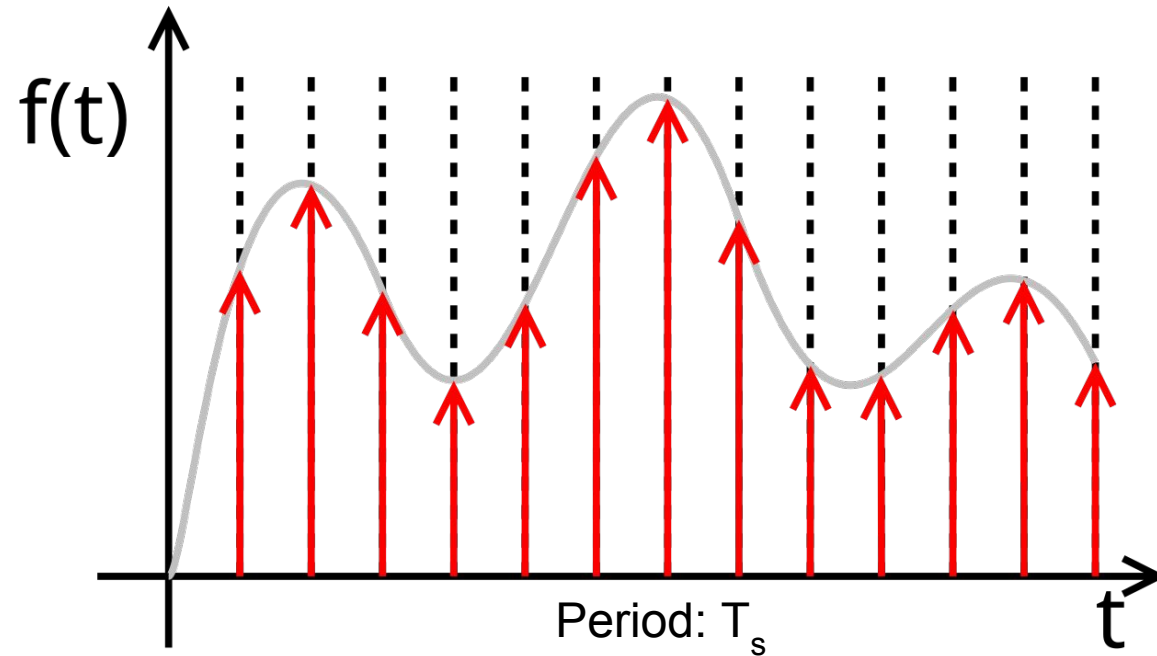
❑ If the signal is already in digital form, no need for analog-to-digital conversion!

Sampling Process



- ❑ Sampling process: Analog signal is converted to discrete through taking samples usually spaced uniformly
- ❑ How to choose sampling rate? Samples should uniquely define the analog signal

Sampling Theorem

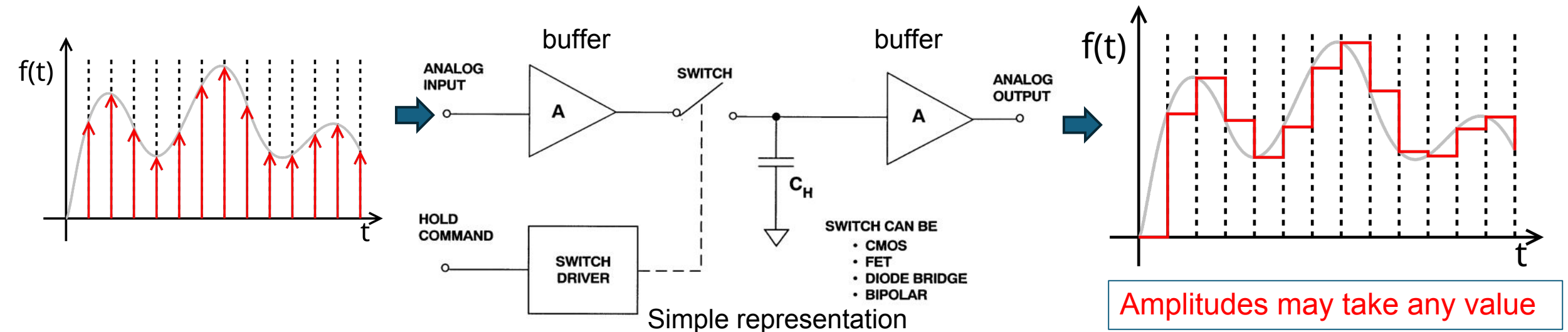


❑ Sampling Theorem: The sampling frequency must be greater than or equal to the twice the maximum frequency component of the signal.

❑ $f_s \geq 2f_m$

❑ $T_s \leq \frac{1}{2f_m}$

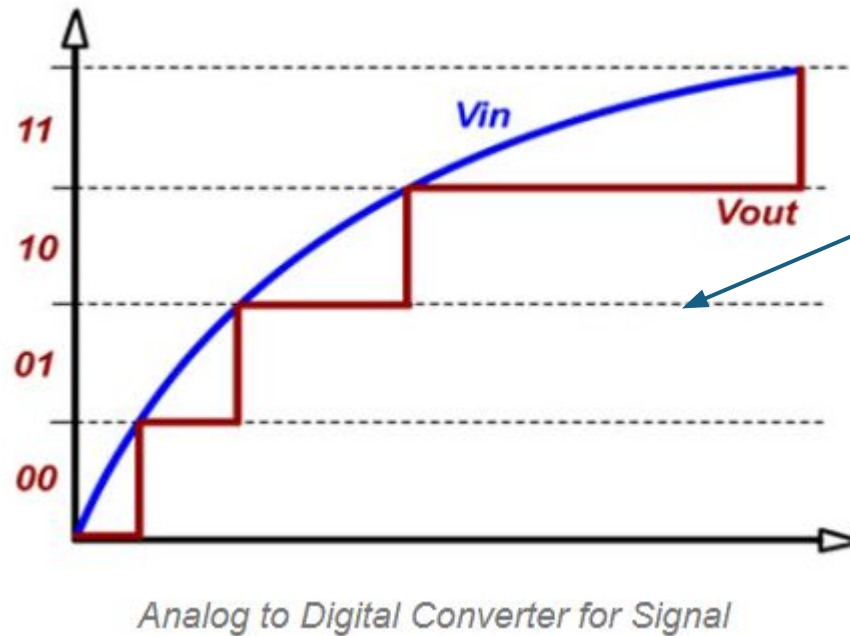
Sample and Hold Circuit



- ❑ Sampling is achieved using a switch opening and closing as per the sampling period
- ❑ The buffer amplifier charges or discharges the capacitor until the voltage across the capacitor matches the input voltage
- ❑ In the track mode, the voltage on the capacitor follows (or tracks) the input signal
- ❑ In hold mode, the switch disconnects the capacitor from the input buffer and output buffer prevents the capacitor from discharging prematurely

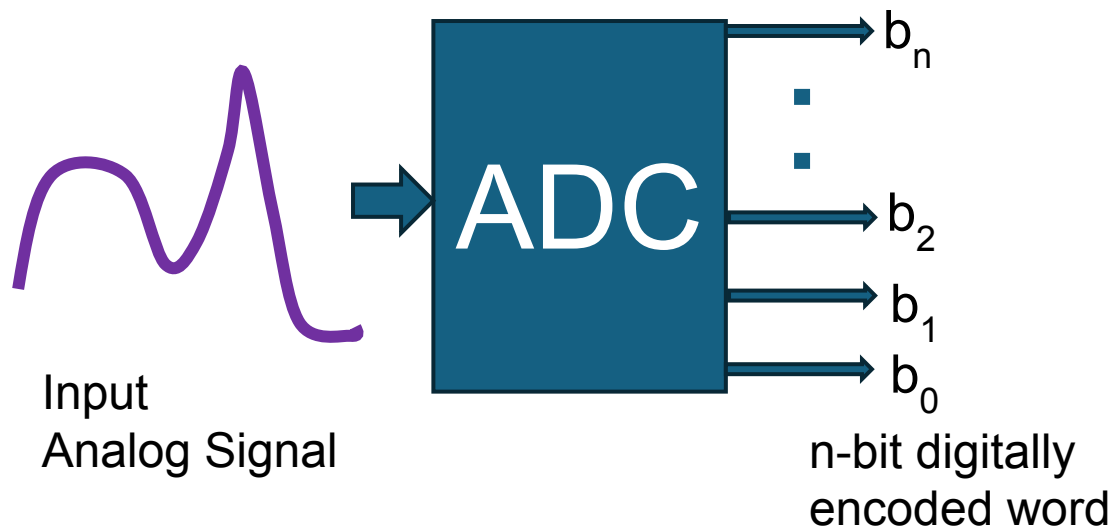
Quantize

In ADC, this is the third block which is mainly used for quantization. The main function of this is to convert the amplitude from continuous (analog) into discrete. The value of continuous amplitude within hold block moves throughout quantize block to turn into discrete in amplitude. Now, the signal will be in digital form because it includes discrete amplitude as well as time.



Fixed amplitudes

Example of 3-bit ADC (Sample, Quantize, Encode)



Output Digital

(d7) b111

(d6) b110

(d5) b101

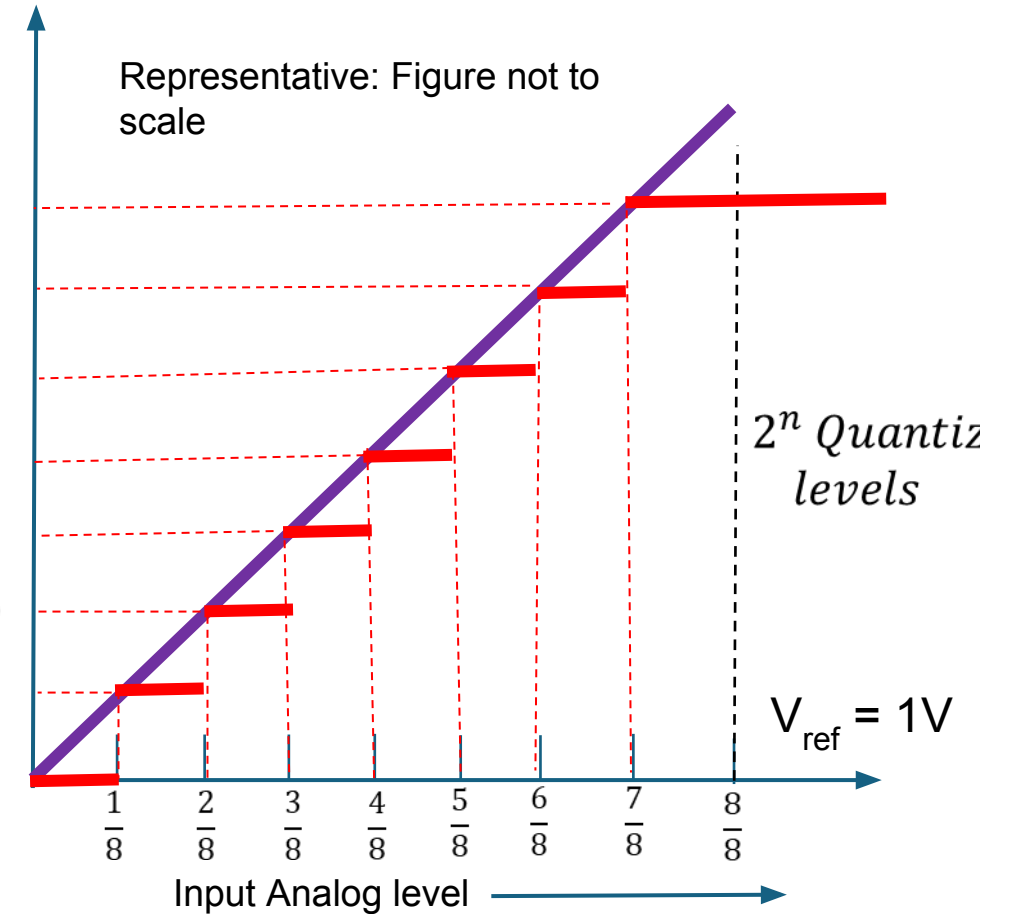
(d4) b100

(d3) b011

(d2) b010

(d1) b001

(d0) b000



Quantization: Rounding off to closest permissible number

$$Resolution = \frac{V_{ref}}{2^n}$$

Example of 3-bit ADC Conversion with 5V Reference

A 3-bit ADC divides the reference voltage (5V) into 8 discrete levels because:

$$2^3 = 8 \text{ (Quantization Levels)}$$

Each level represents a **step size** (resolution), calculated as:

$$\text{Step Size} = \frac{V_{ref}}{2^N} = \frac{5V}{8} = 0.625V$$

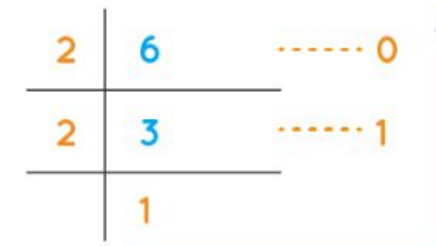
$$\text{Resolution} = \text{step size} = \frac{\text{Reference voltage}}{2^n}$$

$$\text{Digital count} = \left\lfloor \frac{\text{Input voltage}}{\text{Reference voltage}} \right\rfloor \times (2^n)$$

Floor: round to lower integer

$$\text{Digital count} = \text{floor} \left[\frac{\text{Input voltage}}{\text{step size}} \right]$$

Analog Input (V)	Digital count (Decimal)	Binary Output (3-bit)
0.00 - 0.62 V	0	
0.63 - 1.24 V	1	001
1.25 - 1.87 V	2	010
1.88 - 2.49 V	3	011
2.50 - 3.12 V	4	100
3.13 - 3.74 V	5	101
3.75 - 4.37 V	6	110
4.38 - 5.00 V	7	111



$$\therefore 6_{10} = 110_2$$

$$2^2 * b_2 + 2^1 * b_1 + 2^0 * b_0 = \text{Decimal}$$

Example ($V_{\text{ref}} = 5\text{V}$, 3 bits)

$$\text{step size} = \frac{5}{2^3} = \frac{5}{8} = 0.625$$



$$\therefore 6_{10} = 110_2$$



$$V_{\text{in}} = 0.5\text{V} \rightarrow \text{Digital value} = \text{floor} \left[\frac{0.5}{0.625} \right] = 0 \quad \text{Decimal to binary} \rightarrow 000$$

$$V_{\text{in}} = 2.6\text{V} \rightarrow \text{Digital value} = \text{floor} \left[\frac{2.6}{0.625} \right] = 4 \quad \text{Decimal to binary} \rightarrow 100$$

$$V_{\text{in}} = 3.3\text{V} \rightarrow \text{Digital value} = \text{floor} \left[\frac{3.3}{0.625} \right] = 5 \quad \text{Decimal to binary} \rightarrow 101$$

$$V_{\text{in}} = 4.4\text{V} \rightarrow \text{Digital value} = \text{floor} \left[\frac{4.4}{0.625} \right] = 7 \quad \text{Decimal to binary} \rightarrow 111$$

Highest decimal value allowed for 3-bit ADC is 7. Any value above 7 should be floored to

Example ($V_{\text{ref}} = 5V$, 8 bits)

$$\text{step size} = \frac{5}{2^8} = \frac{5}{8} = 0.0195 = 19.5 \text{ mV}$$



$$\therefore 6_{10} = 110_2$$



$$V_{\text{in}} = 0.1V \Rightarrow \text{Digital value} = \text{floor} \left[\frac{100}{19.5} \right] = 5$$

Decimal to
binary

00000101

$$V_{\text{in}} = 2.6V \Rightarrow \text{Digital value} = \text{floor} \left[\frac{2600}{19.5} \right] = 133$$

Decimal to
binary

10000101

$$V_{\text{in}} = 3.3V \Rightarrow \text{Digital value} = \text{floor} \left[\frac{3300}{19.5} \right] = 169$$

Decimal to
binary

10101001

$$V_{\text{in}} = 4.99V \Rightarrow \text{Digital value} = \text{floor} \left[\frac{4990}{19.5} \right] = 255$$

Decimal to
binary

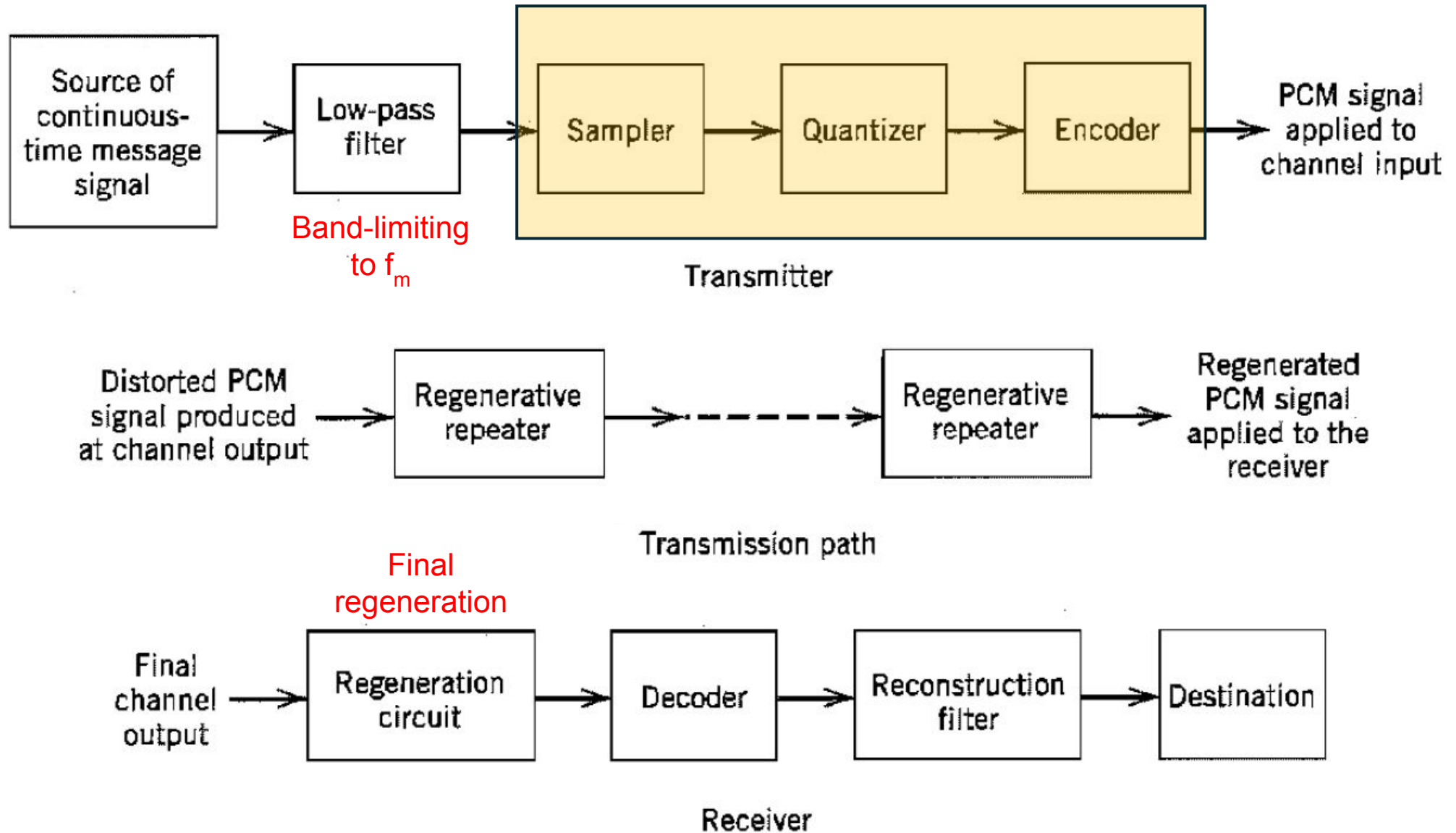
11111111

Highest decimal value allowed for 8-bit ADC is 255. Any value above 255 should be floored to

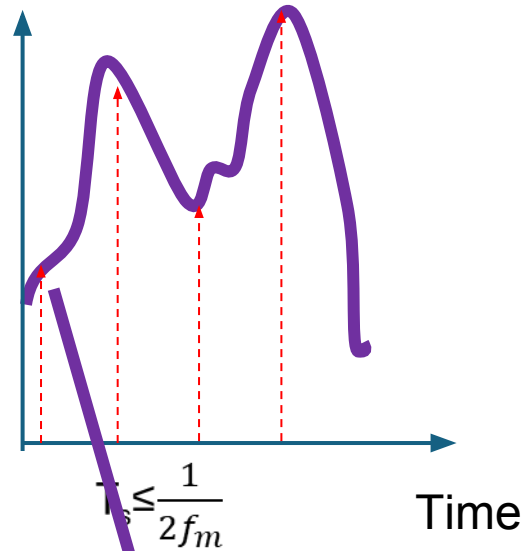
Pulse Coded Modulation

- ❑ PCM is a method of converting Analog signal to digital signal
- ❑ A message is represented by coded pulses
- ❑ **Transmit Process**: sampling -> Quantization -> Encoding
- ❑ **Receive Process**: regeneration -> Decoding-> Reconstruction
- ❑ Not a modulation in traditional sense

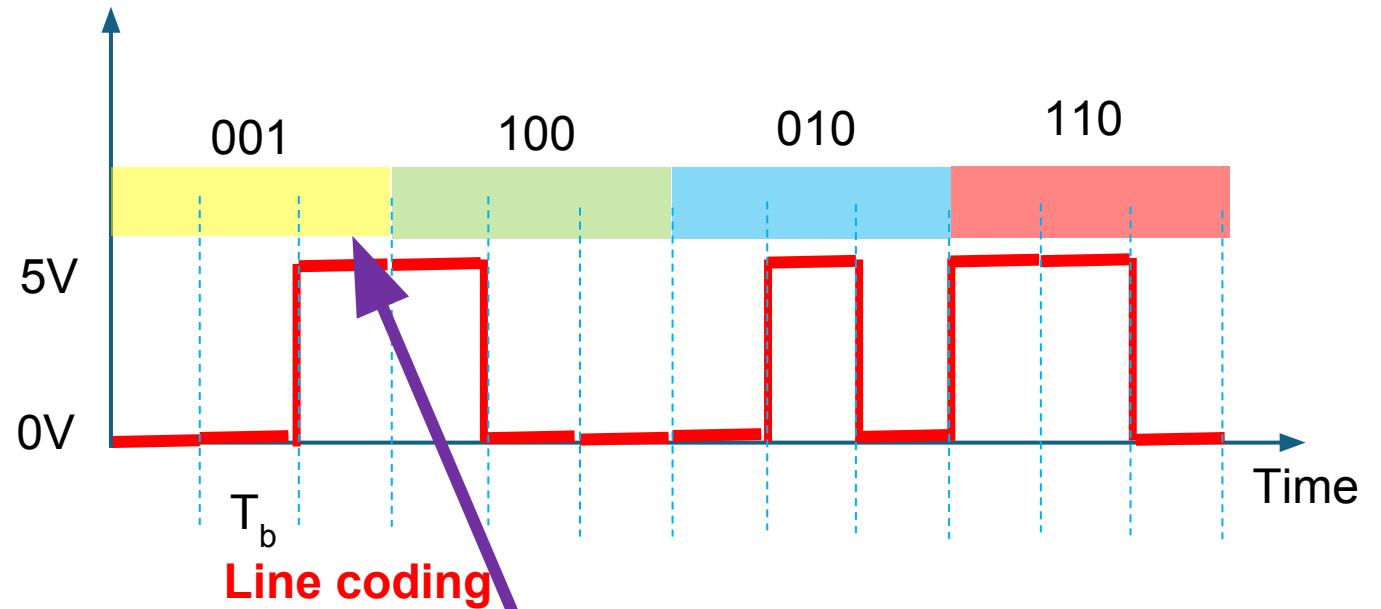
Pulse Coded Modulation



PCM Example



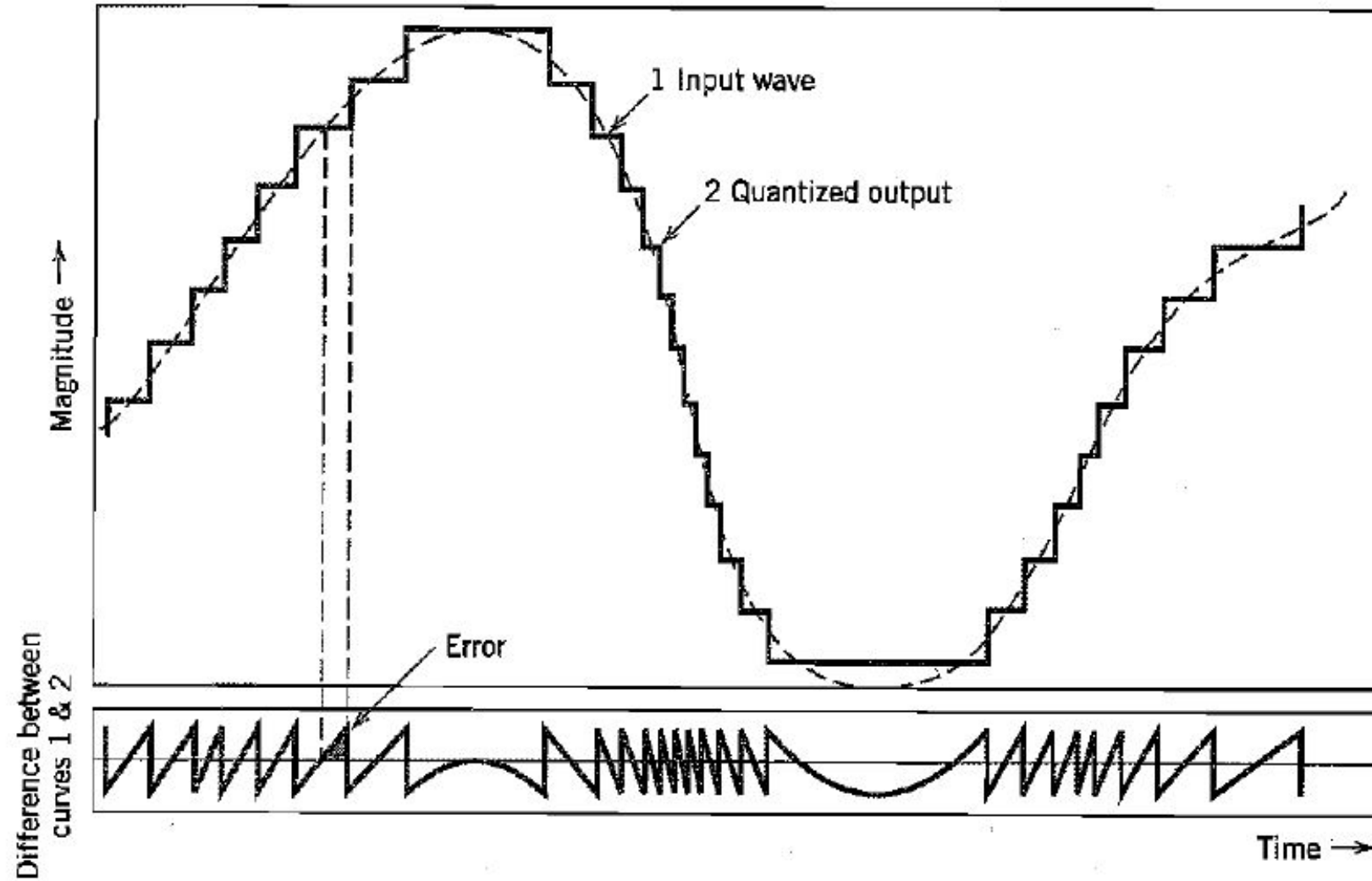
This digital data can be stored (CDs) or transmitted (Telephones)



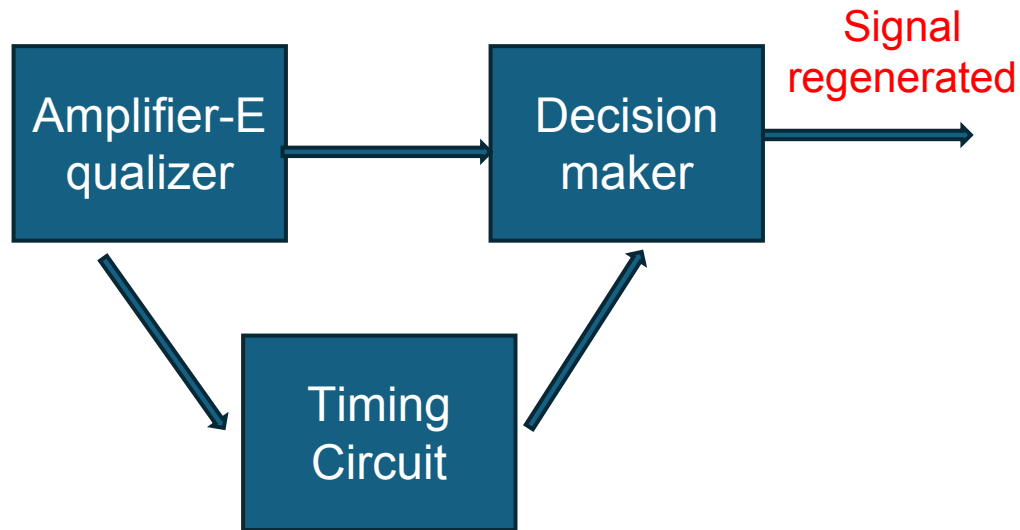
Analog Input (V)	Quantized Level	Binary Code (3-bit)
0.64V	1	001
2.5V	4	100
1.0V	2	010
3.8V	6	110

❑ The sampling frequency must be $\geq$ twice the maximum frequency component of the signal.

Quantization Noise



Regeneration



- ❑ Signal is first amplified and equalized to **remove attenuation and distortion**
- ❑ Timing circuit provides synchronization pulses: **tells when to observe waveform**
- ❑ Decision maker, **compares to a threshold** and **decides** whether bit was either 0 or 1
- ❑ **Bit pattern is again line coded** to get regenerated signal

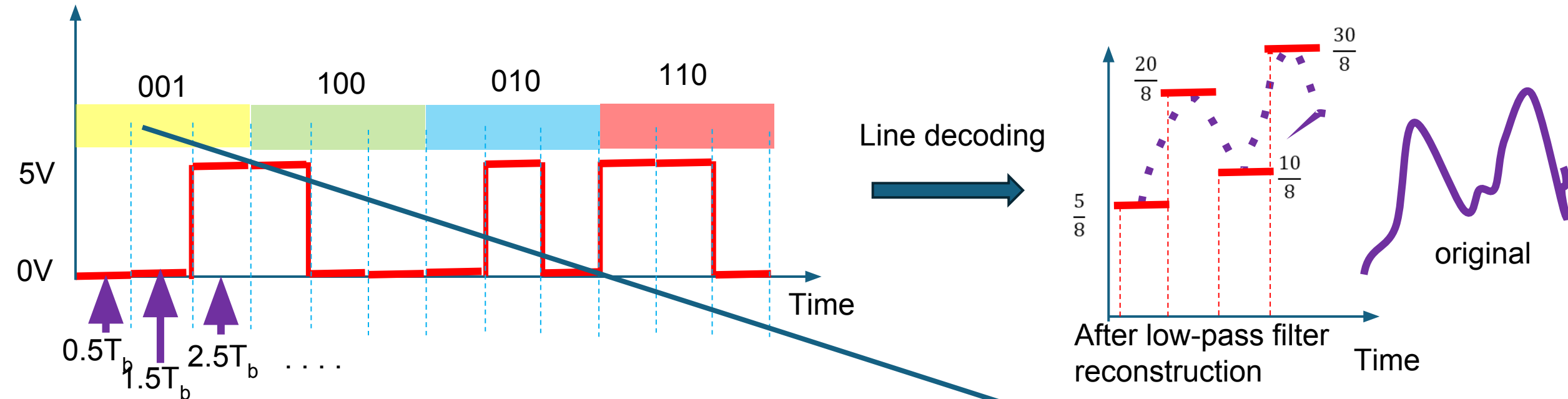
Decoding

- ❑ Follows after final signal clean up and reshaping (regeneration)
- ❑ Pulses are grouped back to code words
- ❑ Code words are mapped back to quantized levels

Filtering

- ❑ Quantized signal is passed through a low-pass filter that smoothens it.
- ❑ Filter cut-off frequency is set to message bandwidth f_m
- ❑ Output contains distortions due to quantization.

PCM Example (Digital to Analog)



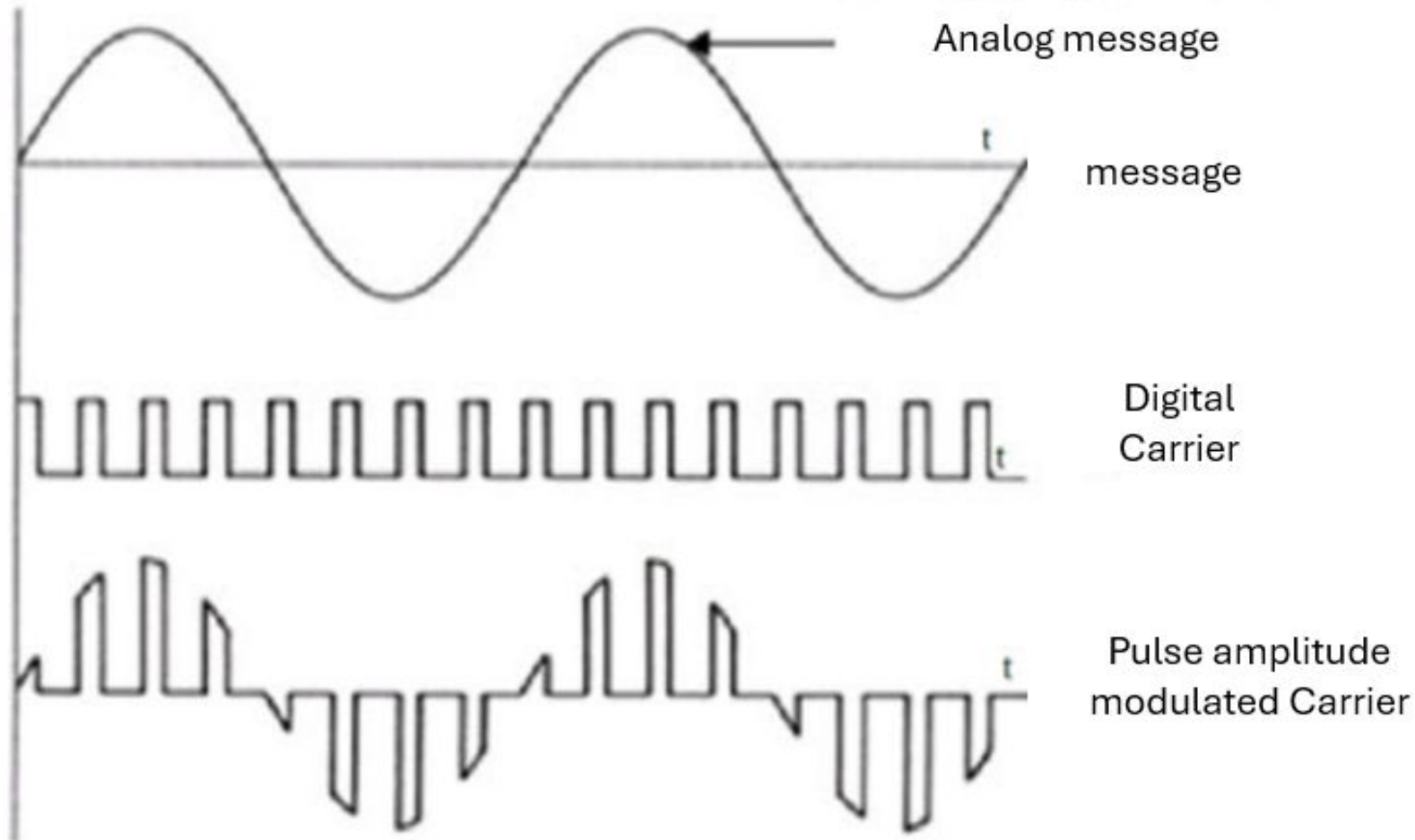
Original (V) (Sent signal)	Analog Output (V) (Estimated signal)	Quantized Level (decimal value)	Binary Code (3-bit)
0.64V	$1 \times 0.625V = 0.625V$	1	001
2.5V	$4 \times 0.625V = 2.5V$	4	100
1.0V	$2 \times 0.625V = 1.25V$	2	010
3.8V	$6 \times 0.625V = 3.75V$	6	110

- ☐ Since while encoding, we used 3 bits together to represent on sample
- ☐ While decoding, we must take 3-bits together and convert from binary to decimal

Other Pulse modulations

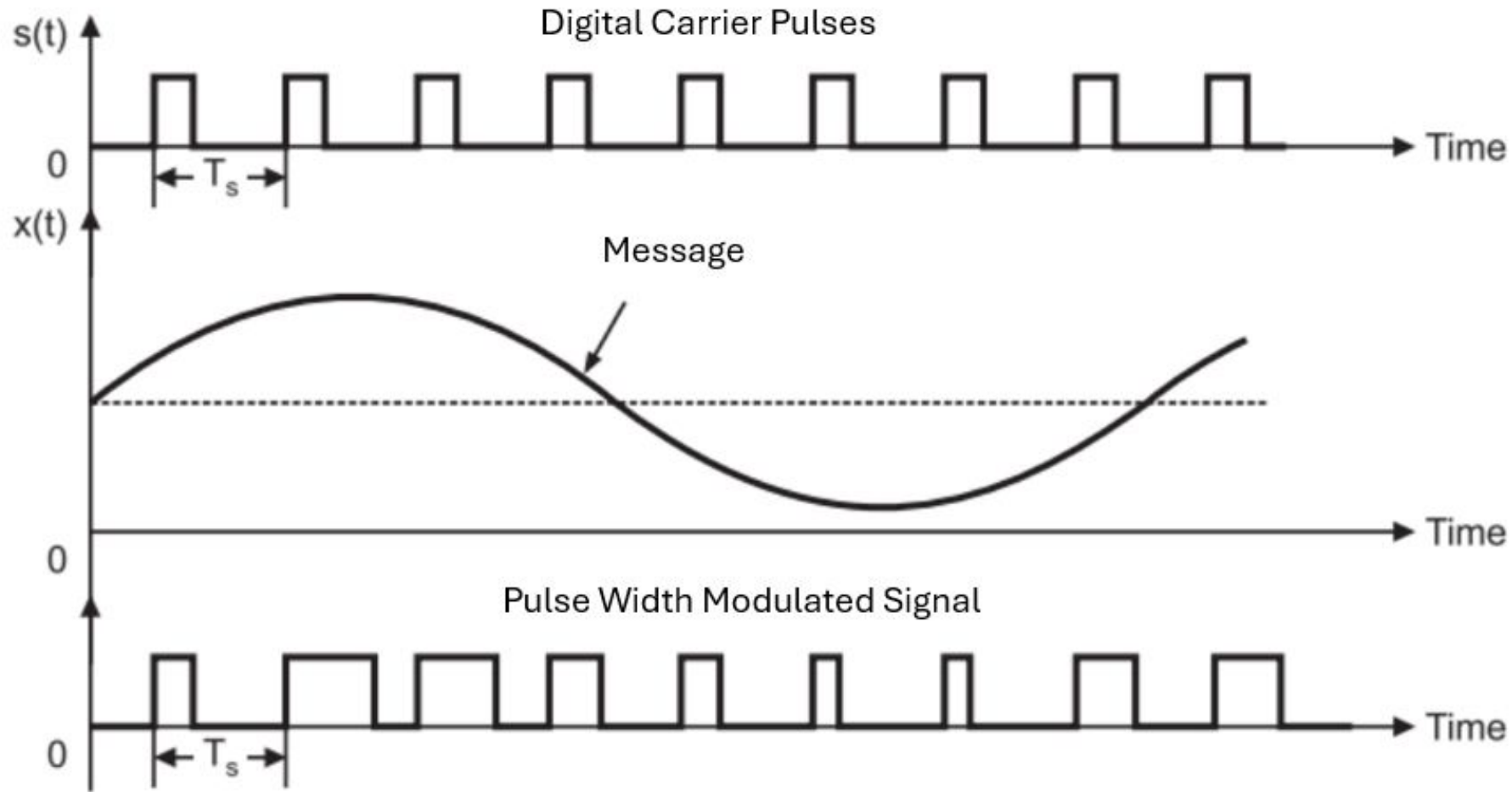
Pulse modulation where some property of a pulse (carrier) is modulated according to analog message signal

Pulse Amplitude Modulation



Amplitude of square wave (digital carrier) changes with analog message signal

Pulse Width Modulation

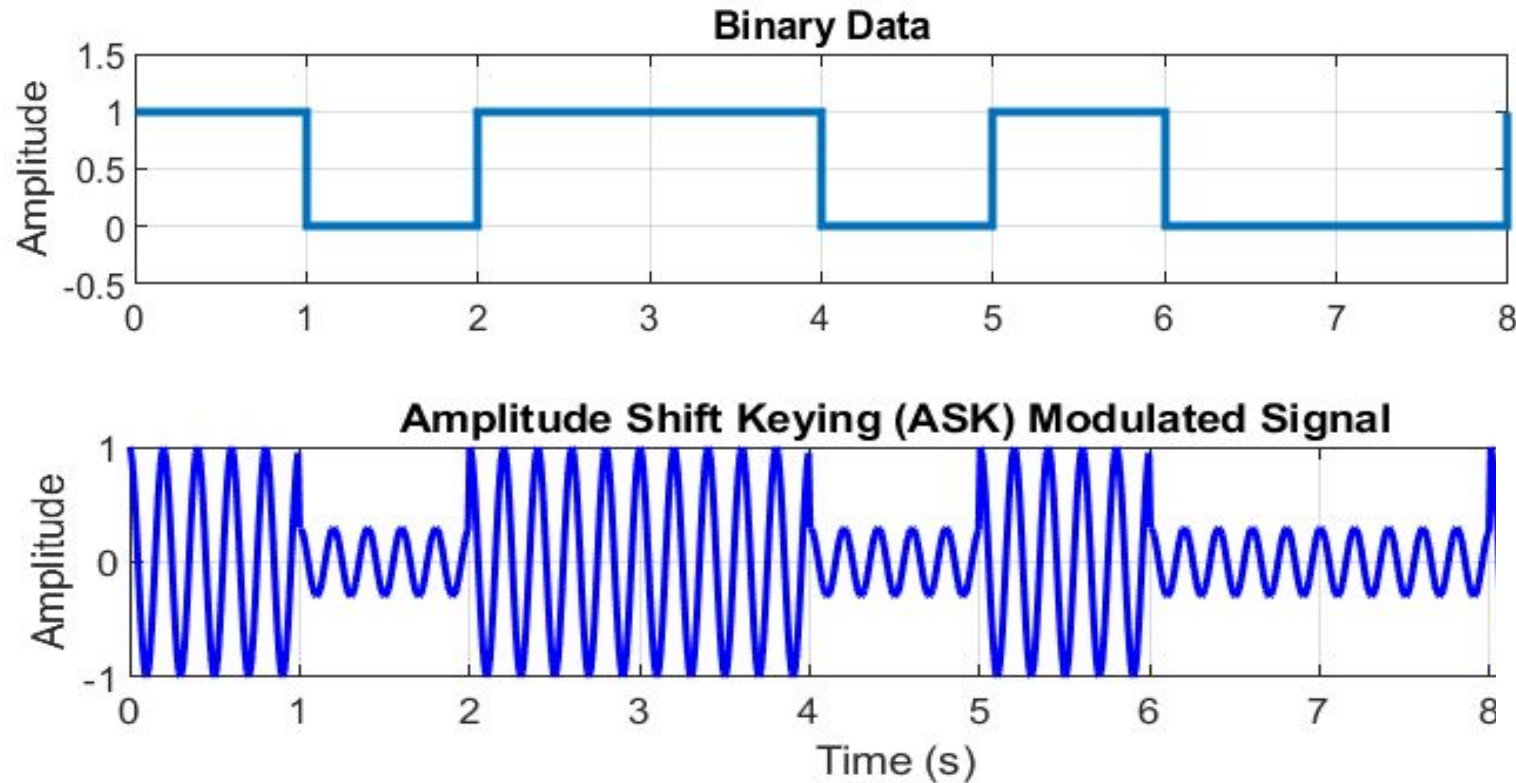


Width of square wave (digital carrier) changes with analog message signal

Digital CW Modulation

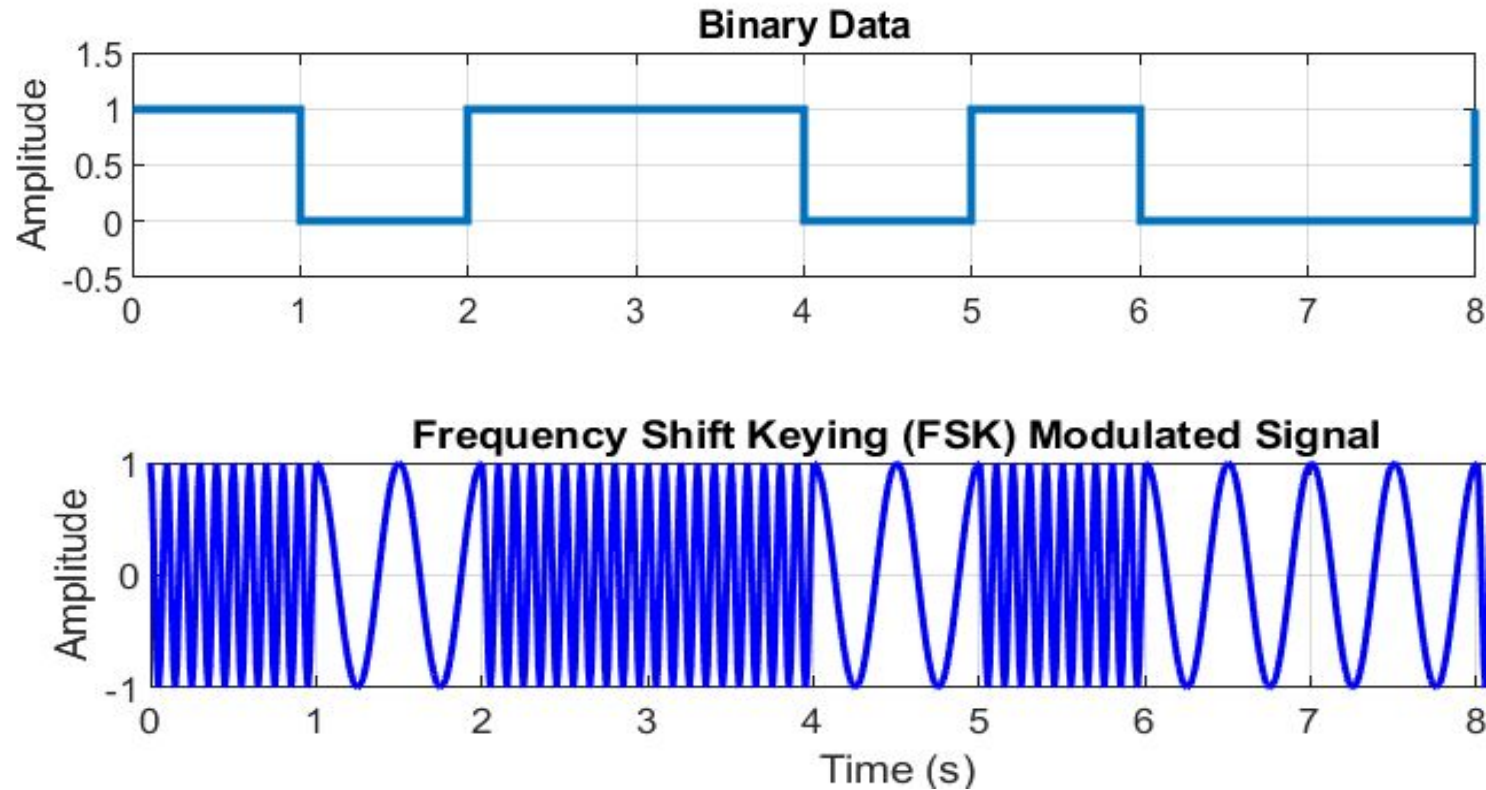
Amplitude shift keying: more suitable technique for digital modulation for wireless channels (passband)

Amplitude shift keying



- ❑ Amplitude of the high frequency sinusoidal carrier wave is varied according to binary data
- ❑ The high frequency sinusoidal carrier is suitable for wireless/fiber/satellite comm. etc. channels in contrast to pulses
- ❑ Here, the carrier is a continuously varying signal in contrast to pulses (hence occupy less bandwidth)

Frequency shift keying



- ❑ Frequency of the high frequency sinusoidal carrier wave is varied according to binary data
- ❑ The envelope of the FSK signal remains constant in contrast to ASK (just like AM/FM case)
- ❑ Here, the carrier is a continuously varying signal in contrast to pulses (hence occupy less bandwidth)

Visible Light communication

- ✓ Laser **Light-Emitting Diodes (LEDs)** are rapidly switched ON and OFF to transmit binary data (1s and 0s).
- ✓ These changes are too fast for the human eye to detect, so the light appears constant.
- ✓ A **photodetector such as light-dependent-resistor** captures the light variations and helps to decode the information.

❑ **Disadvantage: Requires line of sight.**

❑ **Applications:**

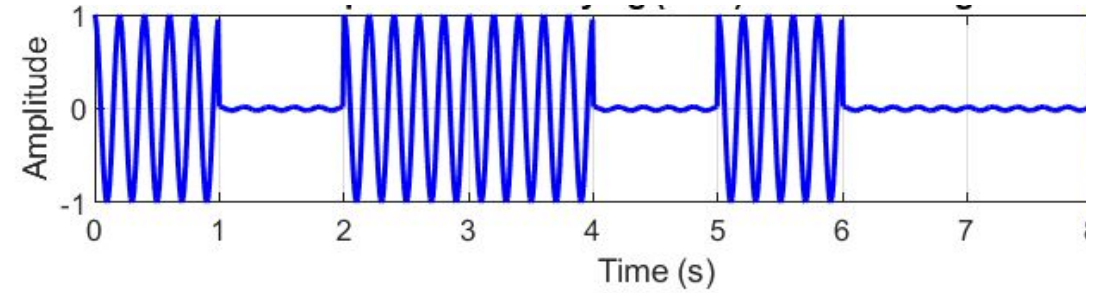
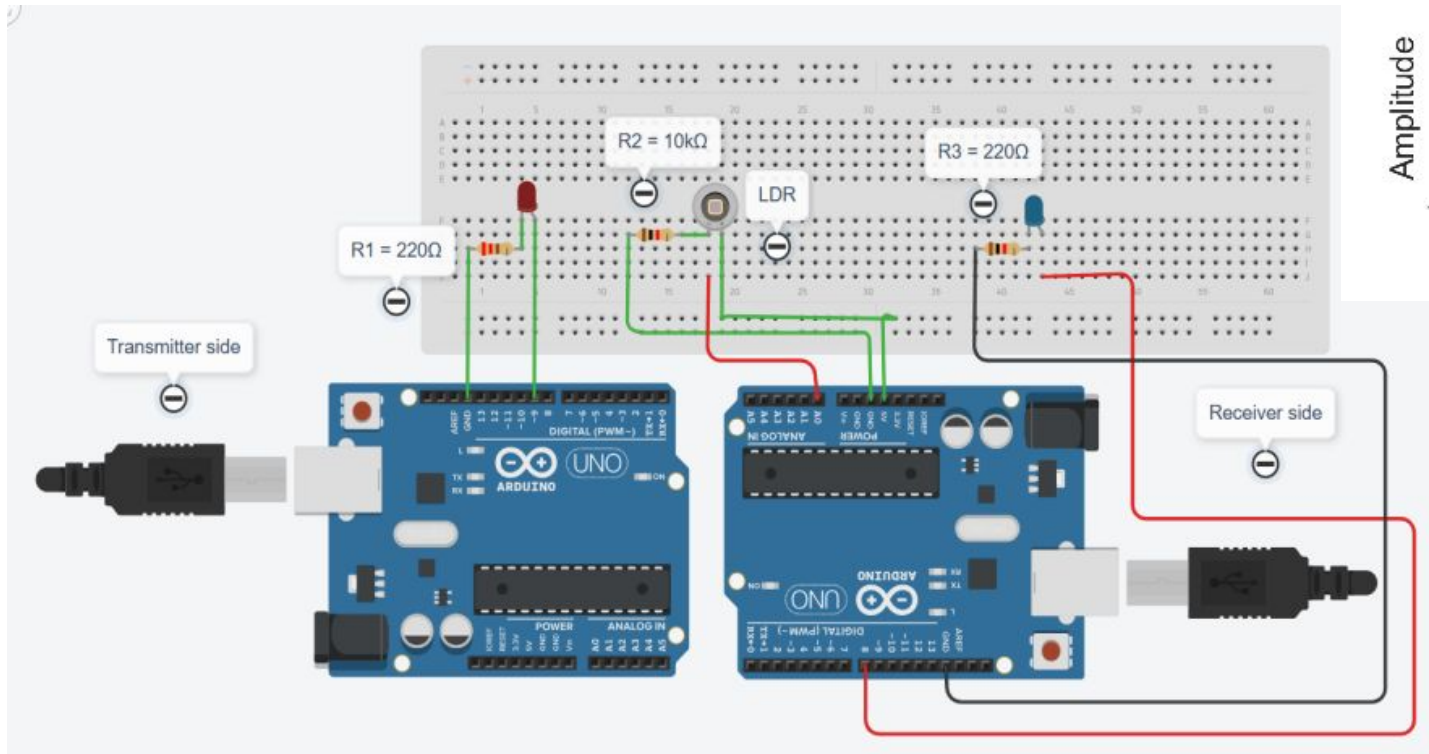
- ❑ **Li-Fi (Light Fidelity):** High-speed internet through LED bulbs.
- ❑ **Underwater Communication:** Efficient alternative to radio waves in marine applications.
- ❑ **Inter-satellite Communications**



Visible Light communication (Lab Experiment)

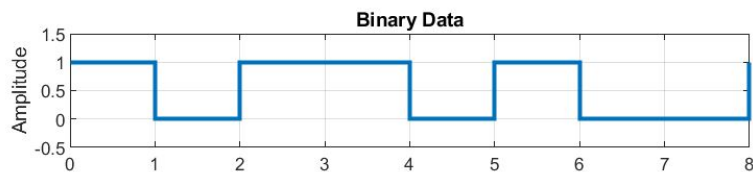
Light signal sent by LED

Light received by LDR

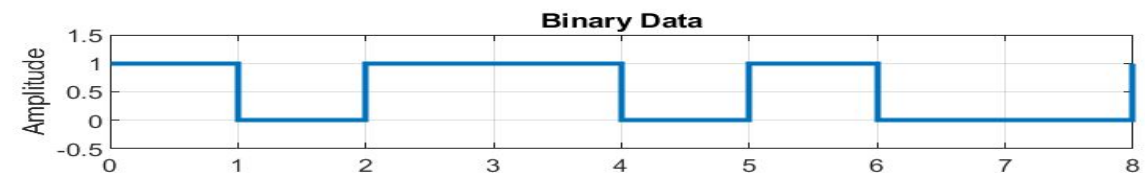


Light Signal is seen as a very high frequency EM wave

In essence, we perform ASK to send the digital data using light as a high frequency carrier



Data to send from Tx Arduino



Data received at Rx Arduino

Conclusion

- ❑ We looked at some of the basic concepts in analog and digital communication techniques
- ❑ We have just touched the surface of the subject
- ❑ Detailed study of these systems will follow along your journey

Wish you all the BEST!

Thank you!